Olalekan 2021 HGSOC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) integrates all 6 patients: same-sample neighbour fraction 0.96→0.38, iLISI median 1.00→3.99 of 6. One connected component (0 islands). Read on the regenerated embedding + per-neighbourhood iLISI, not a scalar: mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

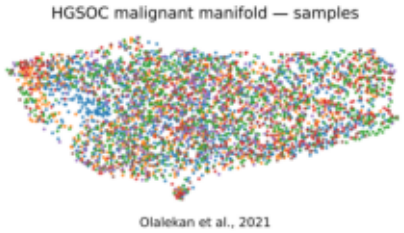
Non-branching case: a single (Gast/pSom)→Crest developmental gradient. Gast & pSom co-vary (smoothed r=+0.80), both oppose Crest (pSom↔Crest r=−0.85, Gast↔Crest r=−0.72). Along pseudotime Gast/pSom fall (ρ −0.77/−0.64), Crest rises to a terminal tip (decile pt=0.99). Joint pSom&Crest enrichment 0.35 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — cross-repression → NA

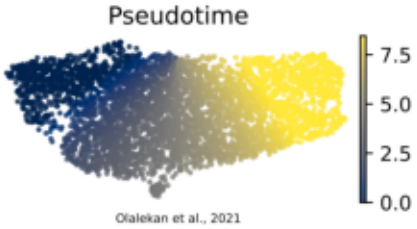
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks are not produced, per the deliverable spec).

STEP FIGURES

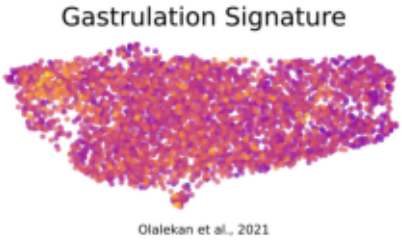
Manifold — samples



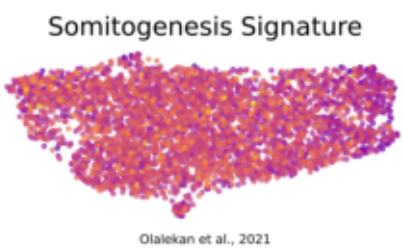
Pseudotime



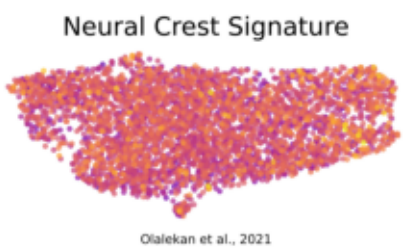
Gastrulation



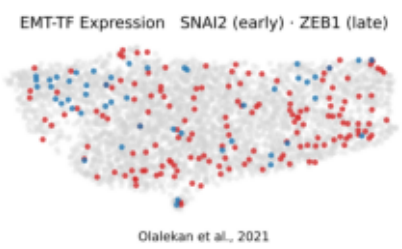
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
HGSOC omental metastases (Olalekan 2021), Drop-seq, 4,125 malignant cells over 6 patients; single low-cycling body (~5% cycling).

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs SNAI2 (early) → ZEB1 (terminus).

Parameter choices

Sample threshold	≥100 cells (no dominant cliff, ratio 1.13); 6/6 samples, 4,125 cells	**Overdispersed genes**	381 (scFates find_overdispersed)
Integration	Harmony on sample, θ=10 (mixing plateau)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 0 islands removed
Geometry	Non-branching; scFates curve N=20, λ=0.3, μ=100 (2 tips / 0 forks)	**Root**	tip 0 = Gast/pSom-high, Crest-low (developmental origin)
Module means (raw)	Gast +0.018 · pSom +0.027 · Crest −0.028 (Linear-pSom/Crest)	**EMT-TFs**	SNAI2 (early/root) ↔ ZEB1 (late/terminus); FOXC/PRRX in panel